

Milestone 4 Report: HotelRevAI – AI-Driven Revenue Analysis for Hotels

1. Project Overview

Problem Statement:

Hotels generate massive amounts of booking data but struggle to extract strategic insights without proper visualization and analytics.

Objectives:

- Analyze revenue performance and growth trends
- Evaluate ADR, cancellation rate, bookings, and guest counts
- Analyze customer behavior and booking patterns
- Build interactive Power BI dashboards
- Enable data-driven hotel management decisions

Dataset Summary:

- Source: Kaggle – Hotel Booking Demand Dataset
- Hotel Types: City Hotel and Resort Hotel
- Records: 75,000+ bookings
- Period: July 2015 – August 2017
- Key Fields: ADR, Lead Time, Customer Type, Distribution Channel, Guests

2. Data Loading & Cleaning

Data Import: Excel dataset imported using Power BI Power Query Editor.

Cleaning Operations:

- Handled missing/null values
- Corrected data types
- Removed duplicate records
- Dropped irrelevant columns

Feature Engineering:

- Total Nights
- Total Guests
- Revenue
- Arrival Date

3. Data Modelling

Tables Created:

- Hotel Data (Fact Table)
- Calendar (Dimension Table)
- D3_Measures (Measure Table)

Relationships: One-to-Many relationship between Calendar[Date] and HotelData[Arrival Date].

4. DAX Measure Library & Business Metrics

Financial Performance Metrics:

- Total Revenue
- Revenue per Night
- Revenue Loss

Customer Behaviour Metrics:

- Total Customers
- Repeat Percentage
- Repeated Guests

Occupancy & Demand Metrics:

- Total Guest Count
- Occupancy Proxy
- Total Guests Stayed

Booking Pattern Metrics:

- Total Bookings
- New Bookings
- Repeated Bookings

Total Measures Created: 16

5. Dashboard 1: Hotel Revenue Performance

- Online Travel Agencies dominate revenue channels
- Transient customers contribute ~74% of revenue
- August is peak revenue month across years

6. Dashboard 2: Customer and Booking Behaviour

- Low loyalty rate (~4.33% repeated guests)
- Portugal leads booking volume (20k+)
- 31–90 day lead time yields strongest performance
- ~70% bookings from transient customers

7. Dashboard 3: Revenue Optimization and Insights

- Demand index and occupancy highest in August
- Advanced bookings generate highest revenue
- Average ADR peaks in August

8. Dashboard 4: Seasonality and Stay Pattern Analysis

- Short stays dominate overall bookings
- 3–4 day stays most frequent
- 4–7 night stays generate ~49% of revenue
- January and November are low-revenue months
- August is consistent peak season

9. Conclusion

The project demonstrates how Power BI transforms raw booking data into actionable insights through data modeling and interactive dashboards. Key revenue drivers include OTA channels, transient customers, August demand peaks, and longer stays. Insights support dynamic pricing, improved cancellation policies, loyalty programs, and direct channel growth.

10. Future Enhancements

- Machine learning demand forecasting
- Real-time booking data pipeline
- Customer segmentation using clustering
- Competitor rate benchmarking
- Automated anomaly detection alerts